

1.Steps

For steps, my heuristic counts how many letters in the current word are different from the goal word.

$h(n)$ = number of letters that are different from the goal

This works because only one letter can be changed in each move. So if there are 3 letters that are different, it will take at least 3 moves to reach the goal. Since every move costs 1, the heuristic will not overestimate the actual remaining cost, so it is admissible.

Scrabble

For Scrabble, I look at each letter that is different from the goal and add the Scrabble value of the goal letter that needs to go in that position.

$h(n)$ = sum of the Scrabble values of the goal letters that still need to be changed

Each incorrect position has to eventually be changed to the correct goal letter, so the cost of inserting those letters has to be paid at some point. If any extra changes are needed along the way, they would only increase the actual cost. Because of this, the heuristic will not overestimate the remaining cost and is admissible.

Frequency

For frequency, I first count the number of letters that are different from the goal. If the current word is already the goal, the heuristic is 0. Otherwise, I use:

$h(n)$ = number of different letters + rarity(goal)

The number of different letters gives the minimum number of moves that still have to be made, and each move has a base cost of at least 1. The solution also has to eventually enter the goal word, so the rarity cost of the goal has to be paid. This means the heuristic only counts costs that are unavoidable, so it does not overestimate the actual remaining cost and is admissible.

2.

Word Pair	Cost Measure	USC Cost	USC Nodes Expanded	A* cost	A* Nodes Expanded	Reduction
WHY → ASK	Steps	5	773	5	16	97.73%

WHY → ASK	Scrabble	9	765	9	13	98.30%
WHY → ASK	Frequency	38.8554	808	38.8554	634	21.53%
BEAR → DUCK	Steps	4	691	4	6	99.13%
BEAR → DUCK	Scrabble	11	2981	11	6	99.89%
BEAR → DUCK	Frequency	38.2402	1084	38.2402	302	72.14%

1. Yes. UCS and A* returned the same optimal cost for all six tests. This shows that A* was able to reduce the number of nodes expanded without changing the optimal solution cost.
2. For WHY → ASK, Scrabble had the largest reduction at 98.30%, and for BEAR → DUCK, Scrabble also had the largest reduction at 99.89%.
3. Frequency was the least informative for both word pairs. For WHY → ASK, it only reduced the nodes expanded by **21.53%**. For BEAR → DUCK, it reduced them by **72.14%**, which was still less than steps and Scrabble.
4. Steps counts the letters that still need to change, while Scrabble also considers the cost of the needed goal letters. Frequency is less informative because it does not account for the rarity costs of intermediate words, so A* may need to expand more nodes.

3.

Steps: Yes

The cost is the same in both directions because every move costs 1. If a path from W1 to W2 takes 3 steps, reversing that path also takes 3 steps. So the optimal cost will be the same.

Scrabble: No

The action can be reversed, but the cost depends on the new letter being added. For example:

dog → cog inserts c, which costs 3.

cog → dog inserts d, which costs 2.

Both words are in the provided dictionary, but the costs are different. So Scrabble cost is not always the same in both directions.

Frequency: No

Frequency cost depends on the word being entered. For example:

dog $\rightarrow$ cog costs $1 + \text{rarity}(\text{cog})$, while
cog $\rightarrow$ dog costs $1 + \text{rarity}(\text{dog})$.

Since these words have different rarity values in the provided dictionary, the costs are different depending on the direction.

Overall, DOGCAT actions are reversible, but that does not mean their costs are always the same in both directions.

4.

Model used: Google Gemini

Exact prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Ai proposed heuristic from gemini:

$h_{\text{rarity}}(n) = h_0(n) \times (1 + R_{\min})$, where $R_{\min}$ is the minimum rarity value of all words in the dictionary.

The idea is that at least $h_0(n)$ moves are needed to reach the goal, and every move costs at least $1 + R_{\min}$.

Yes, it is admissible. At least $h_0(n)$ moves are required because one move can only change one letter. Since every move has a frequency cost of at least $1 + R_{\min}$, the true remaining cost cannot be less than: $h_0(n) \times (1 + R_{\min})$

Therefore, the heuristic does not overestimate the true remaining cost.

It doesn't exactly satisfy the more informative requirement; If $R_{\min} = 0$, then: $h_{\text{rarity}}(n) = h_0(n)$

This shows it would not be strictly larger than h_0 for any state. So even though the proposed heuristic is admissible, it may fail the requirement of being more informative.

For repair, a better heuristic is: $h(n) = 0$, if n is the goal

$h(n) = h_0(n) + \text{rarity}(\text{goal})$, otherwise

This is admissible because at least $h_0(n)$ moves are required, with a base cost of at least 1 per move. The solution must also eventually enter the goal word, so its rarity cost must be paid.

Therefore, this heuristic only includes unavoidable costs and cannot overestimate the true remaining cost.

Testing against h_0

Problem	Heuristic	Solution cost	Nodes expanded
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WHY → ASK	h0	38.8554	755
WHY → ASK	Repaired	38.8554	634
BEAR → DUCK	h0	38.2402	720
BEAR → DUCK	Repaired	38.2402	302

Both heuristics found the same optimal solution costs. The repaired heuristic expanded fewer nodes in both tests. For WHY → ASK, expansions dropped from 755 to 634, and for BEAR → DUCK they dropped from 720 to 302. This shows that adding the unavoidable rarity cost of the goal made the heuristic more informative than h0 while still keeping it admissible.